Agent Sandbox on Cloud Run `preview`

Secure Code Execution for AI Agents

Architecture Security

Key Sandbox CLI Commands

Execution controls and runtime environment configurations

1. **`sandbox do`**
   Executes synchronous operations in an isolated, disposable container environment.

2. **`sandbox run`**
   Run detached, long running process

3. **`sandbox exec`**
   Runs commands inside existing sandbox.

4. **`--mount Flag`**
   Binds filesystems, volumes, or secure directory mounts dynamically across container boundaries.

sandbox_demo.sh

```
$ sandbox do -- python3 script.py
[Disposable] Synchronous execution finished.

$ sandbox run --detach --name=my_bg_srv
Detached server started in background namespace.

$ sandbox exec my_bg_srv -- python3 check.py
[Active Session] Connected & verified.

$ sandbox run --mount=/tmp/host:/mnt/dest
Shared directory mounted with persistent access.
```

Architecture Security

Sandbox Execution Flow

Routing request phases through isolated container namespaces

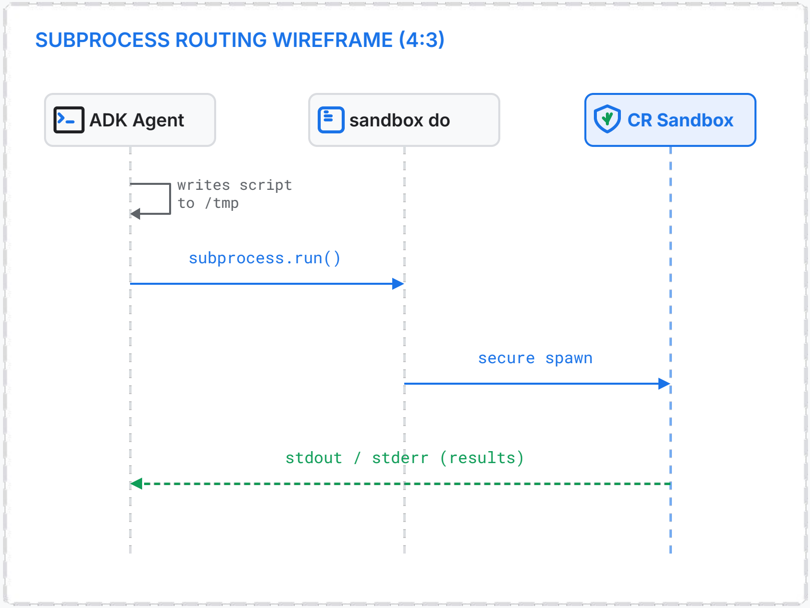
1. **User Request & Routing**
   User sends request to FastAPI App, routing through the ADK Agent to the Gemini Model.

2. **Tool Call Generation**
   Gemini Model processes instructions, generates Python code, and requests a script execution tool call.

3. **Local Script Write**
   ADK Agent triggers the run_python_script tool, writing generated code to /tmp/tmp_code.py.

4. **Subprocess Invocation**
   Tool invokes: ['sandbox', 'do', '--', '/usr/local/bin/python3', '/tmp/tmp_code.py'].

5. **Sandbox Spawning & Return**
   Sandbox Launcher spawns container namespace in Cloud Run Sandbox. Isolation boundaries secure execution; output results flow back to User.



Sandbox Session Management

Unix-Style Execution Tools

Five granular, Unix-style tools manage execution and background sessions within the sandbox environment.

`run_python_script`
Handles Python script blocks within the ephemeral filesystem.

`run_sandbox_command`
Synchronously executes arbitrary CLI commands.

`start_background_sandbox`
Initiates a detached background session for persistent tasks.

`execute_in_background_sandbox`
Runs interactive commands using `sandbox exec` to interact with persistent namespaces.

`stop_background_sandbox`
Terminates and deletes background sessions using `sandbox delete`, ensuring clean lifecycle management.

Sandbox Technical Scenarios

Four distinct scenarios demonstrating isolation, state persistence, background daemons, and secure storage.

Scenario 01

Network Isolation

Local computations succeed while egress network requests fail immediately with DNS errors.

Scenario 02

Session Persistence

Preserves ephemeral state across separate invocations using sync tarball imports and exports.

Scenario 03

Background Daemons

Starts a background daemon web server and executes interactive commands in its namespace.

Scenario 04

Secure Directory Sharing

Enables secure, isolated directory sharing between sandbox containers via session bind mounts.